

# Lección 20

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# Plan de trabajo

## ► Formulación hamiltoniana

– Ecuaciones de Hamilton-Jacobi

## ► Nuestro Syllabus:

UNIDAD 5. Formulación Hamiltoniana

5.1. Transformaciones de Legendre

5.2. Ecuaciones de Hamilton

5.3. Ecuaciones de Hamilton de la Partícula Relativista

5.4. Corchete de Poisson

5.5. Transformaciones Canónicas

5.6. Ecuaciones de Hamilton-Jacobi

5.7. Teoría de perturbaciones

## ► Tarea 2:

– Voy a subir los resultados hoy día

## ► Recordatorio:

– Certamen 2: 03/06/2026

# Hamilton-Jacobi formalism

● Now we are going to discuss briefly another approach to description of dynamics: Hamilton-Jacobi formalism

► It is a straightforward extension of Canonical Formalism

► First we will introduce some auxiliary concept and formulate HJ in general

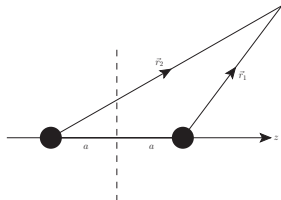
► Next we'll apply it to simplest systems like harmonic oscillator: You will find it too complicated (compared to “conventional” Euler-Lagrange or Newton)

► However, finally, we'll apply it to some potentials which simply cannot be solved analytically using conventional methods, whereas HJ allows to get analytic results (so its complexity pays off).

Some examples: potentials

$$U(r, \theta) = \underbrace{d_0 \frac{\cos \theta}{r^2}}_{\text{electr. dipole}},$$

$$U(r, z) = \underbrace{-\frac{\alpha}{r} + Fz}_{\text{Coulomb+homogeneous field}}$$



Double Coulomb:

$$U(\mathbf{r}) = \frac{\alpha_1}{r_1} + \frac{\alpha_2}{r_2}$$

# Hamilton-Jacobi formalism - main idea

– **Earlier result:** Action is generating functional of canonical transformation

$$(p^{(i)}, q^{(i)}) \rightarrow (p^{(f)}, q^{(f)})$$

(coordinates at moment  $t^{(i)}$  to coordinates at moment  $t^{(f)}$ )

$$\Rightarrow dS = \sum p_a^{(f)} dq_a^{(f)} - \sum p_a^{(i)} dq_a^{(i)} + (H^{(i)} - H^{(f)}) dt$$

$$\frac{\partial S}{\partial q_a^{(f)}} = p_a^{(f)}, \quad \frac{\partial S}{\partial q_a^{(i)}} = -p_a^{(i)}, \quad \frac{\partial S}{\partial t^{(f)}} = -H^{(f)}, \quad \frac{\partial S}{\partial t^{(i)}} = H^{(i)}$$

– From now we'll focus on variations of final coordinates  $q_a^{(f)} \equiv q_a$  and time  $t_2 \equiv t$ :

$$H = H(p, q) = H\left(\frac{\partial S}{\partial q_a}, q_a\right) = -\frac{\partial S}{\partial t}$$

This is a **Hamilton-Jacobi equation** for unknown function  $S$ .

Our goal: find  $S$  and everything associated with it (momenta, canonical transformation, ...)

# Hamilton-Jacobi formalism

## Hamilton-Jacobi equation

$$\frac{\partial S}{\partial t} + H\left(p_a = \frac{\partial S}{\partial q_a}, q_a, t\right) = 0$$

- A *single* equation in partial first-order derivatives.
- Expect that  $S = f(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t) + A$  where  $\alpha_i$ ,  $A = \text{const}$  which must be fixed from initial conditions
- Case of conservative systems:

$$\frac{\partial H}{\partial t} = 0, \quad H(p_{\text{cl}}, q_{\text{cl}}) = E = \text{const}, \quad \Rightarrow E = -\frac{\partial S}{\partial t}$$

$$\Rightarrow S = \underbrace{f(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N)}_{\text{no } t\text{-dependence}} - E t$$

# Jacobi's theorem

If  $S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$  is the solution of the HJ equation, then the system has  $N$  integrals of motion

$$Q_a = \partial S / \partial \alpha_a = \beta_a = \text{const}$$

Formally, this is a system of  $N$  algebraic (not differential) equations, so extraction of trajectory becomes pure algebraic exercise.

**Proof:**

Assume  $S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$  is the solution of HJ equation.

Consider  $S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$  as a generating function  $F_2(q, P, t)$  which depends on old coordinates  $q$  and new momenta  $P$ :

$$p_a = \frac{\partial S}{\partial q_a}, \quad Q_a \equiv \beta_a = \frac{\partial S}{\partial \alpha_a}$$

Since  $S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$  depends on time, need to take into account that Hamiltonian also changes:

$$\tilde{H} = H + \frac{\partial S}{\partial t} = 0 \text{ (in view of HJ equation) (!!!)}$$

## Jacobi's theorem (II)

$$p_a = \frac{\partial S}{\partial q_a}, \quad Q_a \equiv \beta_a = \frac{\partial S}{\partial \alpha_a}$$

$$\tilde{H} = H + \frac{\partial S}{\partial t} = 0 \text{ (in view of HJ equation) (!!!)}$$

Equations of motion in the variables  $(P, Q)$ :

$$\dot{\alpha}_a = -\frac{\partial \tilde{H}}{\partial Q_a} = 0, \quad \dot{\beta}_a = \frac{\partial \tilde{H}}{\partial \alpha_a} = 0,$$

$$\text{Solution :} \quad \Rightarrow \alpha_a = \text{const}, \quad \beta_a = \text{const}.$$

Q.E.D

—Conservation of  $\alpha_a$  was implied from the beginning

\* Constants which determine shape AND invariant under symmetries: energy  $E$ , angular momentum  $M$ ...

—Conservation of  $\beta_a$  is a nontrivial result

\* The  $\beta_a$  are constants which determine orientation in space, time, position ...

\* If your general solution depends on  $t - t_0$ ,  $x - x_0$ ,  $\varphi - \varphi_0$  ... , where  $t_0$ ,  $x_0$ ,  $\varphi_0$ , ... are some constant, then  $\beta_i = \{t_0, x_0, \varphi_0\}$  corresponds to these constants

\*  $S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$  can be interpreted as a generating function of canonical transformation from coordinates  $q_1, \dots, q_N$  to these new “variables”  $\beta_i$

# HJ - summary

## Hamilton-Jacobi equation

$$\frac{\partial S}{\partial t} + H\left(p_a = \frac{\partial S}{\partial q_a}, q_a, t\right) = 0$$

If we found  $S = f(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$ , where  $\alpha_i = \text{const}$  are some constants, can find trajectory from

$$\beta_a = \frac{\partial S}{\partial \alpha_a} = \text{const}, \quad a = 1, \dots, N$$

and get algebraic system of  $N$  coupled equations to extract trajectory (!)

- Earlier we have seen that a single function  $F(\mathbf{q}, \mathbf{Q})$  may encode canonical transformation of  $2N$  variables  $(p_a, q_b) \rightarrow (P_a, Q_b)$
- Now see that a function  $S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$  encodes a full trajectory  $q_a(t)$ , in general  $N$  independent variables

## Control question

– Assume we found  $S = S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$ , where  $\alpha_i = \text{const}$  are some constants, and try find trajectory from

$$\beta_a = \frac{\partial S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)}{\partial \alpha_a} = \text{const}, \quad a = 1, \dots, N$$

[mathematically, this is a local 1-to-1 diffeomorphism  $(\beta_1, \dots, \beta_N) \leftrightarrow (q_1, \dots, q_N)$ .]

– What conditions should satisfy the function  $S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)$  for existence of the solution?

## Control question

$$\beta_a = \frac{\partial S(q_1, \dots, q_N; \alpha_1, \dots, \alpha_N, t)}{\partial \alpha_a} = \text{const}, \quad a = 1, \dots, N$$

**Lemma:** It is possible to construct solution near any point where

$$\det \left( \frac{\partial \beta_a}{\partial q^b} \right) = \det \left( \underbrace{\frac{\partial^2 S}{\partial \alpha^b \partial q^b}}_{\mathcal{M}_{ab}} \right) \neq 0$$

**Proof:** As usual, make linear expansion near some fixed point  $\mathbf{q}^{(0)} = \mathbf{q}_a(t_0)$  assuming smallness of  $\delta q_a = q_a - q_a^{(0)}$ ,

$$\begin{aligned} \beta_a &= \frac{\partial S(q_a^{(0)} + \delta q_a; \alpha_b, t_0 + \delta t)}{\partial \alpha_a} = \underbrace{\frac{\partial S(q_a^{(0)}; \alpha_b, t_0)}{\partial \alpha_a}}_{=\beta_a} + \underbrace{\frac{\partial S(q_a^{(0)}; \alpha_b, t_0)}{\partial \alpha_a \partial q_b}}_{\mathcal{M}_{ab}} \delta q_b + \delta t \frac{\partial S(q_a^{(0)}; \alpha_b, t_0)}{\partial \alpha_a \partial t} \\ &\Rightarrow \delta q_a = -\delta t (\mathcal{M}^{-1})_{ab} \frac{\partial S(q_a^{(0)}; \alpha_b, t_0)}{\partial \alpha_b \partial t} \end{aligned}$$

Solution exists/is uniquely defined if matrix  $\mathcal{M}$  is not degenerate.

## Example: Harmonic oscillator in HJ formalism

Hamilton-Jacobi equation:

$$E = H = -\frac{\partial S}{\partial t} \Rightarrow S = f(q) - E t$$

Explicit hamiltonian:

$$H = \frac{p^2}{2m} + \frac{k q^2}{2} = \frac{1}{2m} \left( \frac{df}{dq} \right)^2 + \frac{k q^2}{2} = E$$

Solve this ODE:

$$\frac{df}{dq} = \sqrt{2m \left( E - \frac{k q^2}{2} \right)}$$

$$\begin{aligned} \Rightarrow f(q) &= \int dq \sqrt{2m \left( E - \frac{k q^2}{2} \right)} = \\ &= E \sqrt{\frac{m}{k}} \arctan \left( \frac{q \sqrt{k}}{\sqrt{2E - k q^2}} \right) + \frac{q}{2} \sqrt{m (2E - k q^2)} + S_0 \end{aligned}$$

## Example: Harmonic oscillator in HJ formalism

$$\begin{aligned} S &= f(q) - E t = \\ &= E \sqrt{\frac{m}{k}} \arctan \left( \frac{q \sqrt{k}}{\sqrt{2E - k q^2}} \right) + \frac{q}{2} \sqrt{m (2E - k q^2)} - E t + S_0 \end{aligned}$$

• Identify free constants:  $\left( \underbrace{E}_{\alpha}, S_0 \right)$

$$\Rightarrow \frac{\partial S}{\partial \alpha} = \underbrace{\sqrt{\frac{m}{k}}}_{1/\omega} \arctan \left( \frac{\sqrt{k q^2}}{\sqrt{2E - k q^2}} \right) - t = 0 \quad \Rightarrow \sqrt{\frac{k q^2}{2E - k q^2}} = \tan(\omega t)$$

$$1 + \tan^2 z = \frac{1}{\cos^2 z} \Rightarrow \cos^2 z = \frac{1}{1 + \tan^2 z}$$

$$\frac{2E - k q^2}{2E} = \cos^2 \omega t \Rightarrow q(t) = \sqrt{\frac{2E}{k}} \sin \omega t$$

in agreement with expectations.

# Motion in non-central field

- For harmonic oscillator application of HJ formalism is definitely overkill, it is much easier to solve (Euler-Lagrange) equations directly
- Now we'll consider more complicated case, when HJ is the only way to proceed

## Example

The charged particle of mass  $m$  moves in the field of electric dipole, so the potential of interaction in spherical coordinates is given by

$$U(r, \theta) = U_0 \frac{\cos \theta}{r^2}.$$

- Identify what symmetries has action of this particle, and what integrals of motion correspond to these symmetries.
- Integrate the equations of motion and find the trajectory of the particle

(we'll see a bit later how to apply this method to even more general potentials).

# Analysis in Lagrange formalism

- Lagrangian in spherical coordinates:

$$L = \frac{m}{2} \left( \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2 \right) - U_0 \frac{\cos \theta}{r^2}$$

## Integrals of motion:

- Energy:

$$E = \frac{m}{2} \left( \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2 \right) + U_0 \frac{\cos \theta}{r^2}$$

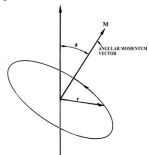
- projection of angular momentum onto axis of dipole

$$M_z = m r^2 \sin^2 \theta \dot{\phi}$$

# Analysis in Lagrange formalism

▷  $\mathbf{M}_\perp \neq \text{const}$ , so in general trajectory does not lie in a plane. Recall that in central field conservation of momentum was vital to prove that trajectory is plane:

Theorem from Lecture 7: If  $\mathbf{M} = \text{const}$ , then:



● Particle always moves in a plane ortogonal to  $\mathbf{M}$ , i.e. if in the moment  $t_0$  vectors  $(\mathbf{r}, \mathbf{v})$  lie in a plane ortogonal to  $\mathbf{M}$ ,

then in central field they remain in this plane for  $\forall t$ :

$$\vec{r}(t_0 + \Delta t) \approx \vec{r}(t_0) + \Delta t \vec{v}, \quad \vec{r}(t_0) \cdot \vec{M} = 0$$

$$\vec{v}(t_0 + \Delta t) \approx \vec{v}(t_0) + \Delta t \vec{a}, \quad \vec{v}(t_0) \cdot \vec{M} = 0$$

$$\Delta t \vec{v} \cdot \vec{M} = \Delta t \vec{v} \cdot m(\vec{r} \times \vec{v}) = 0,$$

$$\Delta t \vec{a} \cdot \vec{M} = \frac{\Delta t \vec{F} \cdot \vec{M}}{m} = \frac{\Delta t \vec{r} \cdot \vec{M}}{m} \frac{dU}{r dr} = 0$$

$\Rightarrow \vec{r}(t_0 + \Delta t), \vec{v}(t_0 + \Delta t)$  will be also ortogonal to  $\vec{M}$

# Analysis in Lagrange formalism

► Full Euler-Lagrange equations of motion (Lecture 7):

$$m\ddot{r} \equiv m \frac{d^2 r}{dt^2} = \frac{2U_0}{r^3} \cos \theta + mr \left( \dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 \right) \quad (1)$$

$$m r^2 \ddot{\theta} + 2mr \dot{r} \dot{\theta} - \frac{m r^2 \sin 2\theta \dot{\phi}^2}{2} = \frac{U_0}{r^3} \sin \theta \quad (2)$$

$$\frac{d(m r^2 \sin^2 \theta \dot{\phi})}{dt} = m r^2 \sin^2 \theta \ddot{\phi} + \\ + m r^2 \sin 2\theta \dot{\theta} \dot{\phi} + 2m r \dot{r} \dot{\phi} \sin^2 \theta = 0 \quad (3)$$

► After substitution

$$\dot{\phi} \rightarrow \frac{M_z}{m r^2 \sin^2 \theta}$$

end up with coupled 2nd order ODEs (1, 2) for  $r(t)$ ,  $\theta(t)$ , which can be solved only numerically.

# Analysis in Hamilton-Jacobi formalism

- System is conservative, so on trajectories which are solutions of equations of motion

$$H = E = \text{const}$$

$$\Rightarrow S = f(r, \theta, \phi) - E t \quad (1)$$

$$\text{Hamiltonian :} \quad H = \frac{\mathbf{p}^2}{2m} + U_0 \frac{\cos \theta}{r^2} = E \quad (2)$$

In spherical coordinates:

$$\mathbf{p} = \nabla S = \left( \frac{\partial S}{\partial r}, \frac{1}{r} \frac{\partial S}{\partial \theta}, \frac{1}{r \sin \theta} \frac{\partial S}{\partial \phi} \right) \quad (3)$$

$$(2, 3) \Rightarrow \frac{1}{2m} \left( \frac{\partial S}{\partial r} \right)^2 + \frac{1}{2mr^2} \left( \frac{\partial S}{\partial \theta} \right)^2 + \frac{1}{2mr^2 \sin^2 \theta} \left( \frac{\partial S}{\partial \phi} \right)^2 + U_0 \frac{\cos \theta}{r^2} = E$$

- We ended with nonlinear system of equation in Partial Derivatives. We need to recall how to solve it mathematically.

*Please let me know if You have any ideas or suggestions*

## HJ: separation of variables

- Assume that the equation in partial derivatives was reduced to the form which allows to isolate a variable  $q_1$  and its derivatives  $\partial S / \partial q_1$

$$0 = \Phi \left( \phi \left( q_1, \frac{\partial S}{\partial q_1} \right), \frac{\partial S}{\partial q_s}, q_s, t \right), \quad s > 1$$

—We may try to find the solution in a separated form

$$S = f(q_s; \alpha_s) + S_1(q_1)$$

$$\underbrace{\phi \left( q_1, \frac{\partial S}{\partial q_1} \right)}_{\text{only } q_1} = \alpha_1 = \text{const}, \quad \underbrace{\Phi \left( \alpha_1, \frac{\partial S}{\partial q_s}, q_s, t \right)}_{s-1 \text{ variables, all except } q_1} = 0$$

- If  $q_1$  is cyclic variable, then  $\phi$  does not depend on  $q_1$  at all, so may get solution:

$$S = \alpha_1 q_1 + S(q_s, \alpha_s)$$

- If we can successively separate all the variables, integration becomes trivial, and

$$S = S_1(q_1) + S_2(q_2) + S_3(q_3) + \dots$$

# Don't confuse

For linear ODEs (e.g. Laplace's equation) separation of variables usually implies search of solutions in the form  $S = f_1(q_1) \cdot \dots \cdot f_N(q_N)$

In HJ such ansatz usually doesn't work due to completely different structure (though idea is similar, we try to reduce equation to a sum of contributions which depend on independent variables)

# Claim

It is possible to integrate equations of motion using the separation of variables for the particle of mass  $m$  moving in the field

$$U(r, \theta, \phi) = a(r) + \frac{b(\theta)}{r^2} + \frac{c(\phi)}{r^2 \sin^2 \theta}$$

where  $a, b, c$  are arbitrary functions.

## HJ: separation of variables (example)

- System is conservative,  $H = E = \text{const}$

$$\Rightarrow S = f(r, \theta, \phi) - E t$$

$$H = \frac{p^2}{2m} + U = E$$

In spherical coordinates:

$$p = \nabla S = \left( \frac{\partial f}{\partial r}, \frac{1}{r} \frac{\partial f}{\partial \theta}, \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \right)$$

$$2m(E - U) = p^2 = \left( \frac{\partial f}{\partial r} \right)^2 + \frac{1}{r^2} \left[ \left( \frac{\partial f}{\partial \theta} \right)^2 + \frac{1}{\sin^2 \theta} \left( \frac{\partial f}{\partial \phi} \right)^2 \right]$$

Look for solution in separable form:

$$f(r, \theta, \phi) = f_3(\phi) + f_2(\theta) + f_1(r)$$

## HJ: separation of variables (example)

$$2m(E - U) = p^2 = \left(\frac{\partial f}{\partial r}\right)^2 + \frac{1}{r^2} \left[ \left(\frac{\partial f}{\partial \theta}\right)^2 + \frac{1}{\sin^2 \theta} \left(\frac{\partial f}{\partial \phi}\right)^2 \right]$$

Look for solution in separable form:  $f(r, \theta, \phi) = f_3(\phi) + f_2(\theta) + f_1(r)$

$$\begin{aligned} & \left( E - a(r) - \frac{1}{2m} \left( \frac{df_1}{dr} \right)^2 \right) + \frac{1}{r^2} \left( b(\theta) - \frac{1}{2m} \left( \frac{df_2}{d\theta} \right)^2 \right) + \\ & + \frac{1}{r^2 \sin^2 \theta} \left( \underbrace{c(\phi) - \frac{1}{2m} \left( \frac{df_3}{d\phi} \right)^2}_{=\text{const}=\alpha_3} \right) = 0 \end{aligned}$$

•

$$\Rightarrow f_3(\phi) = \int d\phi \sqrt{2m(c(\phi) - \alpha_3)}$$

• if  $c(\phi) = 0$ , then  $\phi$  is cyclic and  $f_3(\phi) = -p_\phi \phi = \phi \sqrt{-\alpha_3}$

## HJ: separation of variables (example)

$$\left( E - a(r) - \frac{1}{2m} \left( \frac{df_1}{dr} \right)^2 \right) + \frac{1}{r^2} \underbrace{\left( b(\theta) - \frac{1}{2m} \left( \frac{df_2}{d\theta} \right)^2 + \frac{\alpha_3}{\sin^2 \theta} \right)}_{=\text{const}=\alpha_2} = 0$$

$$\Rightarrow f_2(\theta) = \int d\theta \sqrt{2m \left( b(\theta) + \frac{\alpha_3}{\sin^2 \theta} - \alpha_2 \right)}$$

$$\Rightarrow E - a(r) - \frac{1}{2m} \left( \frac{df_1}{dr} \right)^2 + \frac{\alpha_2}{r^2} = 0$$

$$\Rightarrow f_1(r) = \int dr \sqrt{2m \left( E - a(r) + \frac{\alpha_2}{r^2} \right)}$$

● Free constants:  $(E, \alpha_2, \alpha_3)$

## HJ: separation of variables (example)

Action:

$$S = -Et + \underbrace{\int d\phi \sqrt{2m (c(\phi) - \alpha_3)}}_{f_3(\phi)} + \underbrace{\int d\theta \sqrt{2m \left( b(\theta) + \frac{\alpha_3}{\sin^2 \theta} - \alpha_2 \right)}}_{f_2(\theta)} \\ + \underbrace{\int dr \sqrt{2m \left( E - a(r) + \frac{\alpha_2}{r^2} \right)}}_{f_1(r)}$$

● Free constants:  $(\alpha_1 \equiv E, \alpha_2, \alpha_3)$

–To get trajectory from  $S$ , we need to take derivatives

$$\frac{\partial S}{\partial \alpha_i} = \text{const} = \beta_i$$

# HJ: separation of variables (example)

Equations for trajectory

$$\frac{\partial f_1(r, E)}{\partial E} - t = \text{const} = -t_0,$$
$$\Rightarrow 2m \int \frac{dr}{\sqrt{2m(E - a(r) - \frac{\alpha_2}{r^2})}} = t - t_0$$

$\Rightarrow$  Gives  $r(t)$ -dependence. Result for  $r(t)$  coincides with we got in Lecture 7 in central potential  $U(r)$  if  $\alpha_2 \equiv M^2/2m$ . However now  $\mathbf{M} \neq \text{const}$ .

$$\frac{\partial S(r, E)}{\partial \alpha_2} = \beta_2 = \text{const},$$

$$\int^r \frac{\frac{dr}{r^2}}{\sqrt{2m(E - a(r) + \frac{\alpha_2}{r^2})}} - \int^\theta \frac{d\theta}{\sqrt{2m(b(\theta) + \frac{\alpha_3}{\sin^2 \theta} - \alpha_2)}} = \beta_2$$

Allows to relate  $\theta$  to variable  $r$  and thus find its time dependence  $\theta = \theta(r(t)) = \theta(t)$

–Finally, the last equation  $\partial S/\partial \alpha_3 = \beta_1 = \text{const}$  allows to relate  $\theta$  and  $\phi$

## Special case

Let's consider central field ( $b = c = 0$ ) and assume for simplicity that  $\alpha_3 = 0$  (motion in plane  $\phi = \text{const}$ )

$$\Rightarrow f_2(\theta) = \int d\theta \sqrt{2m \left( b(\theta) + \frac{\alpha_3}{\sin^2 \theta} - \alpha_2 \right)} = \sqrt{2m\alpha_2} i (\theta - \theta_0)$$

$\Rightarrow \partial S(r, E) / \partial \alpha_2$  will yield familiar

$$\theta - \theta_0 = \int^r dr \frac{M}{m r^2 \sqrt{\frac{2}{m} (E - U_{\text{eff}}(r))}} \quad (3)$$

in agreement with our earlier result from Lecture 7.

—If we do not assume  $\alpha_3 = 0$  and use

$$\frac{\partial S}{\partial \alpha_2} = 0, \quad \frac{\partial S}{\partial \alpha_3} = 0$$

we would get equivalent result, but dependence on  $\theta, \phi$  is much more complicated:  
it is motion in a plane *inclined* with respect to axis  $Z$

# HJ formalism - dipole field

Special case: field of dipole,  $b(\theta) = \cos \theta$ ,  $a(r) = c(\phi) = 0$

$$\Rightarrow f_3(\phi) = \int d\phi \sqrt{2m(c(\phi) - \alpha_3)} = \sqrt{2m\alpha_3}\phi$$

$$\Rightarrow f_2(\theta) = \int d\theta \sqrt{2m\left(-\cos \theta + \frac{\alpha_3}{\sin^2 \theta} - \alpha_2\right)}$$

$$\Rightarrow f_1(r) = \int dr \sqrt{2m\left(E - \frac{\alpha_2}{r^2}\right)}$$

• Free constants:  $(\alpha_1 \equiv E, \alpha_2, \alpha_3)$

$$\frac{\partial f_1(r, E)}{\partial E} - t = \text{const} = -t_0,$$

## HJ formalism - dipole field

$$\frac{\partial f_1(r, E)}{\partial E} - t = \text{const} = -t_0 \Rightarrow 2m \int \frac{dr}{\sqrt{2m(E - \frac{\alpha_2}{r^2})}} = t - t_0$$

$$r^2(t) = \frac{E^2(t - t_0)^2 + 2\alpha_2 m}{2Em}$$

- Particle falls on center if  $\alpha_2 < 0$ , scatters otherwise
- General case complicated, not in plane.
- ... Mathematica allows to take integrals (express them via some special (elliptic) functions) in general case, but result is lengthy. For brevity we'll consider  $\alpha_3 = 0$  (corresponds to motion in plane  $\phi = \text{const}$ ), possible only if symmetry axis  $\hat{z}$  belongs to the plane formed by  $\mathbf{r}(0)$  and  $\dot{\mathbf{r}}(0)$ .

$$f_2(\theta) = \sqrt{2m(\alpha_2 - 1)}E \left( \frac{\theta}{2} \middle| \frac{2}{1 - \alpha_2} \right)$$

$$\frac{\partial S(r, E)}{\partial \alpha_2} = \beta_2 = \text{const}, \text{ gives } \theta = \theta(r) :$$

$$-\frac{1}{\sin\left(r\sqrt{E/\alpha_2}\right)\sqrt{2m\alpha_2}} - \sqrt{\frac{2}{m(\alpha_2 - 1)}} F\left(\frac{\theta}{2} \middle| \frac{2}{1 - \alpha_2}\right) = \beta_2$$

- General case is NOT motion in plane. We can get algebraic relations (NOT ODE) which formally allow to get trajectory

## HJ: separation of variables

Note that full separation of variables in a given system of coordinates is possible only for a certain class of potentials, which we may “guess” from structure of  $(\nabla S)^2$ .

Analyze the most general form of potential which allows a full separation of variables in Hamilton-Jacobi method in cylindrical coordinates

$$\rho = \sqrt{x^2 + y^2}, \quad \tan \phi = y/x$$
$$\left(\frac{\partial S}{\partial \rho}\right)^2 + \frac{1}{\rho^2} \left(\frac{\partial S}{\partial \phi}\right)^2 + \left(\frac{\partial S}{\partial z}\right)^2 = 2m(E - U) \quad (1)$$

## HJ: separation of variables

$$\left(\frac{\partial S}{\partial \rho}\right)^2 + \frac{1}{\rho^2} \left(\frac{\partial S}{\partial \phi}\right)^2 + \left(\frac{\partial S}{\partial z}\right)^2 = 2m(E - U) \quad (1)$$

$$S = f_1(\rho) + f_2(\phi) + f_3(z)$$

$$\left(\frac{\partial f_1(\rho)}{\partial \rho}\right)^2 + \frac{1}{\rho^2} \left(\frac{\partial f_2(\phi)}{\partial \phi}\right)^2 + \left(\frac{\partial f_3(z)}{\partial z}\right)^2 = 2m(E - U) \quad (1)$$

We can see that derivatives

$$\frac{\partial S}{\partial \rho}, \frac{\partial S}{\partial \phi}, \frac{\partial S}{\partial z}$$

are functions of only  $\rho$ ,  $\phi$ ,  $z$  respectively

$\Rightarrow$  from (1) conclude that separation of variables is possible if the potential in right-hand side has a dependence on variables  $\rho$ ,  $\phi$ ,  $z$  similar to the left-hand side:

$$U = a(\rho) + \frac{b(\phi)}{\rho^2} + c(z)$$

# Conditions for separation of variables in HJ formalism

The separation is possible if the following conditions are satisfied:

## Staeckel conditions

- $\partial H / \partial t = 0$
- Hamiltonian has quadratic form

$$H = \frac{1}{2} (p_i - a_i) (T^{-1})_{ij} (p_j - a_j) + V(q)$$

- Vectors  $a_i$  depend only on respective coordinate,  $a_i = a_i(q_i)$
- The potential might be written as

$$V(q) = \sum \frac{V_i(q_i)}{T_{ii}}$$

- Assume there is a matrix  $\phi$  such that

$$\delta_{ij} \phi_{ij}^{-1} = \frac{1}{T_{ii}}, \quad \frac{\partial S}{\partial q_i} - a_i = 2\delta_{ik} \phi_{kj} \gamma_j$$

where  $\gamma_j$  is a constant vector. Then  $\phi_{ii}$  and  $(\phi^{-1})_{ii}$  depend only on variable  $q_i$

In many cases easier to substitute & check directly ...

## Example 2

The particle moves in the field

$$U = -\frac{\alpha}{r} + Fz$$

(a combination of the Coulomb and homogeneous external field) . Integrate the equations of motion and find the trajectories of the particle

- Can't use cylindrical coordinates, potential includes term

$$-\frac{\alpha}{\sqrt{\rho^2 + z^2}}$$

(not  $a(\rho) + b(\phi)/\rho^2 + c(z)$ )

- Can't use spherical coordinates, potential includes term

$$Fz = Fr \cos \theta$$

(not  $b(\theta)/r^2$ )

## Example 2

The particle moves in the field

$$U = -\frac{\alpha}{r} + Fz$$

(a combination of the Coulomb and homogeneous external field) . Integrate the equations of motion and find the trajectories of the particle

**Help:** You may find useful parabolic coordinate system

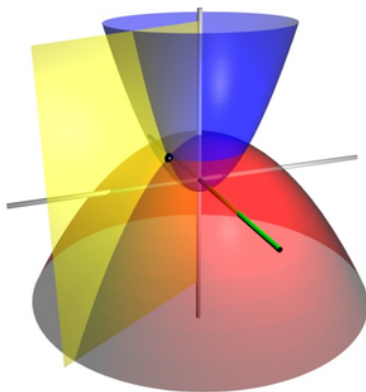
$$z = \frac{\xi - \eta}{2}, \quad \rho = \sqrt{\xi\eta} = \sqrt{x^2 + y^2},$$

$$r = \sqrt{\rho^2 + z^2}$$

$$\xi = r + z, \quad \eta = r - z$$

$$z = \frac{\xi - \eta}{2}, \quad r = \frac{\xi + \eta}{2}$$

$$x = \sqrt{\xi\eta} \cos \phi, \quad y = \sqrt{\xi\eta} \sin \phi$$



## Example 2

Lagrangian in parabolic coordinate system

$$L = \frac{m}{2} (\dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2) - U = \frac{m(\xi + \eta)}{8} \left( \frac{\dot{\xi}^2}{\xi} + \frac{\dot{\eta}^2}{\eta} \right) + \frac{m\xi\eta\dot{\phi}^2}{2} - U$$

$$p_{\xi} = \frac{m}{4} \frac{\xi + \eta}{\xi} \dot{\xi}, \quad p_{\eta} = \frac{m}{4} \frac{\xi + \eta}{\eta} \dot{\eta}, \quad p_{\phi} = m\xi\eta\dot{\phi}$$

$$H = \frac{2}{m} \frac{\xi p_{\xi}^2 + \eta p_{\eta}^2}{\xi + \eta} + \frac{p_{\phi}^2}{2m\xi\eta} + U$$

We'll show that separation of variables works for general potential

$$U = \frac{a(\xi) + b(\eta)}{\xi + \eta}$$

(our case corresponds to  $a(\xi) = \alpha - \frac{1}{2}F\xi^2$ ,  $b(\eta) = \alpha + \frac{1}{2}F\eta^2$ )

## Example 2

Seek solutions in the form:

$$S = f_1(\xi) + f_2(\eta) + f_3(\phi) - E t$$

•  $p_\phi = \text{const}$  since  $\phi$  is cyclic variable  $\Rightarrow f_3 = p_\phi \phi$

$$H = \frac{2}{m} \frac{\xi (f'_1(\xi))^2 + \eta (f'_2(\eta))^2}{\xi + \eta} + \frac{p_\phi^2}{2m\xi\eta} + \frac{a(\xi) + b(\eta)}{\xi + \eta} = E$$

After multiplication by  $\xi + \eta$  we end up with (below  $F_{1,2}$  are known functions)

$$\left[ \xi (f'_1(\xi))^2 + F_1(\xi) \right] + \left[ \eta (f'_2(\eta))^2 + F_2(\eta) \right] = 0$$

The first term depends only on  $\xi$ , second only on  $\eta$ , so sum=0 only if each of them is constant ( $= \pm \alpha_1$ )

$$2\xi (f'_1(\xi))^2 + \frac{p_\phi^2}{2\xi} - m(E\xi - a(\xi)) = \alpha_1$$

$$2\eta (f'_2(\eta))^2 + \frac{p_\phi^2}{2\eta} - m(E\eta - b(\eta)) = -\alpha_1$$

## Example 2

**Formal solution:**

$$f_1(\xi) = \int^\xi \frac{d\xi}{\sqrt{2\xi}} \sqrt{\alpha_1 - \frac{p_\phi^2}{2\xi} + m(E\xi - a(\xi))}$$

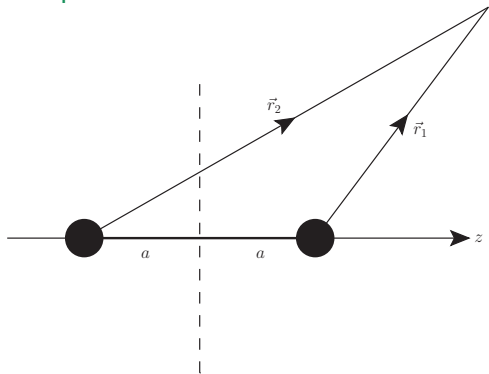
$$f_2(\eta) = \int^\eta \frac{d\eta}{\sqrt{2\eta}} \sqrt{-\alpha_1 - \frac{p_\phi^2}{2\eta} + m(E\eta - b(\eta))}$$

• Free constants:  $\alpha_k = (\alpha_1, E, p_\phi)$ . Request  $\partial S / \partial \alpha_k = \beta_k = \text{const}$  to get trajectories.

– Our initial problem corresponds to  $a(\xi) = \alpha - \frac{1}{2}F\xi^2$ ,  $b(\eta) = \alpha + \frac{1}{2}F\eta^2$ .

\*Integration with *Mathematica*:  $f_{1,2} \rightarrow$  elliptic functions (so end up with implicit trajectory)

### Example 3



A charged particle of mass  $m$  moves in the Coulomb field of two fixed charges located at distance  $2a$ ,

$$U = \frac{\alpha_1}{r_1} + \frac{\alpha_2}{r_2}$$

where  $\alpha_{1,2} = \text{const}$  and  $r_{1,2} \equiv |\vec{r}_{1,2}|$  are distances from fixed points, as shown in the plot. Integrate the equations of motion for this particle.

## Example 3

► We'll use elliptic coordinates  $(\xi, \eta, \phi)$ , which are related to the polar coordinates  $(\rho, z, \phi)$  as

$$\rho = a\sqrt{(\xi^2 - 1)(1 - \eta^2)}, \quad z = a\xi\eta.$$

In these coordinates the distances  $r_{1,2} \equiv \sqrt{(z \mp a)^2 + \rho^2}$  shown in the Figure might be rewritten as

$$r_1 = a(\xi - \eta), \quad r_2 = a(\xi + \eta)$$

► Lagrangian:

$$\begin{aligned} L = T - U &= \frac{m}{2} \left( \dot{\rho}^2 + \rho^2 \dot{\phi}^2 + \dot{z}^2 \right) - U = \\ &= \frac{ma^2}{2} (\xi^2 - \eta^2) \left( \frac{\dot{\xi}^2}{\xi^2 - 1} + \frac{\dot{\eta}^2}{1 - \eta^2} \right) + \frac{ma^2}{2} (\xi^2 - 1)(1 - \eta^2) \dot{\phi}^2 - \frac{\alpha_1/a}{\xi - \eta} - \frac{\alpha_2/a}{\xi + \eta} \end{aligned}$$

## Example 3

► Generalized momenta:

$$p_\xi = \frac{\partial L}{\partial \dot{\xi}} = m a^2 \left( \frac{\xi^2 - \eta^2}{\xi^2 - 1} \right) \dot{\xi}, \quad p_\eta = \frac{\partial L}{\partial \dot{\eta}} = m a^2 \left( \frac{\xi^2 - \eta^2}{1 - \eta^2} \right) \dot{\eta},$$

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \frac{m a^2}{2} (\xi^2 - 1) (1 - \eta^2) \dot{\phi}$$

► Use Legendre transform to get Hamiltonian:

$$\begin{aligned} H &= \dot{\xi} p_\xi + \dot{\eta} p_\eta + \dot{\phi} p_\phi - L = \\ &= \frac{(\xi^2 - 1)p_\xi^2 + (1 - \eta^2)p_\eta^2}{2ma^2(\xi^2 - \eta^2)} + \frac{p_\phi^2}{2ma^2(\xi^2 - 1)(1 - \eta^2)} + \frac{\alpha_1/a}{\xi - \eta} + \frac{\alpha_2/a}{\xi + \eta} \end{aligned}$$

## Example 3

► Hamilton-Jacobi equations:

$$\frac{\partial S}{\partial t} + H \left( p_\xi \equiv \frac{\partial S}{\partial \xi}, p_\eta \equiv \frac{\partial S}{\partial \eta}, p_\phi \equiv \frac{\partial S}{\partial \phi} \right) = 0,$$

$\Downarrow$

$$\frac{\partial S}{\partial t} + \frac{(\xi^2 - 1)(\partial_\xi S)^2 + (1 - \eta^2)(\partial_\eta S)^2}{2ma^2(\xi^2 - \eta^2)} + \frac{(\partial_\phi S)^2}{2ma^2(\xi^2 - 1)(1 - \eta^2)} + \frac{\alpha_1/a}{\xi - \eta} + \frac{\alpha_2/a}{\xi + \eta} = 0,$$

►  $\phi$  is cyclic variable,  $p_\phi = \text{const}$

$$S = -Et + p_\phi \phi + F(\xi, \eta) \quad (1)$$

► apply separation of variables (assume that  $F(\xi, \eta) = f_1(\xi) + f_2(\eta)$ ):

$$S = -Et + p_\phi \phi + f_1(\xi) + f_2(\eta). \quad (2)$$

## Example 3

► HJ after substitution:

$$E = \frac{(p_\phi)^2}{2ma^2 (\xi^2 - 1) (1 - \eta^2)} + \frac{(\xi^2 - 1) (\partial_\xi f_1(\xi))^2 + (1 - \eta^2) (\partial_\eta f_2(\eta))^2}{2ma^2 (\xi^2 - \eta^2)} + \frac{(\alpha_1 + \alpha_2) \xi + (\alpha_1 - \alpha_2) \eta}{a(\xi^2 - \eta^2)},$$

► Looks like dependence on  $\xi$ ,  $\eta$  is not separable, but in reality it is:

- multiply both parts by  $\xi^2 - \eta^2$
- take into account

$$\frac{\xi^2 - \eta^2}{(\xi^2 - 1) (1 - \eta^2)} = \frac{1}{\xi^2 - 1} + \frac{1}{1 - \eta^2}$$

$$E (\xi^2 - \eta^2) = \frac{(\xi^2 - 1) (\partial_\xi f_1(\xi))^2}{2ma^2} + \frac{(1 - \eta^2) (\partial_\eta f_2(\eta))^2}{2ma^2} + \frac{p_\phi^2}{2ma^2} \left( \frac{1}{\xi^2 - 1} + \frac{1}{1 - \eta^2} \right) + \frac{(\alpha_1 + \alpha_2) \xi + (\alpha_1 - \alpha_2) \eta}{a},$$

– sum of the terms which depend either on  $\xi$  (in red) or on  $\eta$  (in blue).

- Put all  $\xi$ - and  $\eta$ -dependent variables in different parts of equation request that both should be a const

## Example 3

$$-E \xi^2 + \frac{(\xi^2 - 1) (\partial_\xi f_1(\xi))^2}{2ma^2} + \frac{p_\phi^2}{2ma^2} \frac{1}{\xi^2 - 1} + \frac{(\alpha_1 + \alpha_2) \xi}{a} = \text{const} = \beta$$

$$E \eta^2 + \frac{(1 - \eta^2) (\partial_\eta f_2(\eta))^2}{2ma^2} + \frac{p_\phi^2}{2ma^2} \frac{1}{1 - \eta^2} + \frac{(\alpha_1 - \alpha_2) \eta}{a} = -\beta = \text{const.}$$

–1st order “ODEs” for functions  $f_1(\xi)$ ,  $f_2(\xi)$ , integration straightforward:

–... extract  $\partial_\xi f_1(\xi)$  and  $\partial_\eta f_2(\eta)$ , after that take integrals

$$S(\xi, \eta, \phi, t) = -E t + p_\phi \phi +$$

$$\int^\xi d\xi \left[ \frac{2ma^2}{\xi^2 - 1} \left( E \xi^2 - \frac{p_\phi^2}{2ma^2} \frac{1}{\xi^2 - 1} - \frac{(\alpha_1 + \alpha_2) \xi}{a} + \beta \right) \right]^{1/2}$$

$$+ \int^\eta d\eta \left[ \frac{2ma^2}{1 - \eta^2} \left( -E \eta^2 - \frac{p_\phi^2}{2ma^2} \frac{1}{1 - \eta^2} - \frac{(\alpha_1 - \alpha_2) \eta}{a} - \beta \right) \right]^{1/2}$$

►Extract trajectory as usual, taking derivatives over free parameters:

$$\frac{\partial S}{\partial \beta} = \text{const}, \quad \frac{\partial S}{\partial p_\phi} = \text{const}, \quad \frac{\partial S}{\partial E} = \text{const}.$$

# Summary

- Hamilton-Jacobi formalism:

- ▶ Canonical transformation to “new” variables, where:

- ★ hamiltonian  $\tilde{H} = 0$
- ★ New canonical variables  $Q_a \equiv \alpha_a = \text{const}$  and  $P_a = \beta_a = \text{const}$
- ★ Main equation of motion:

$$H = H(p, q) = H\left(\frac{\partial S}{\partial q_a}, q_a\right) = -\frac{\partial S}{\partial t}$$

- ▶ Powerful tool:

- ★ Too sophisticated for simple problems
- ★ Helps to solve some nontrivial problems
- ★ “Standard” method: separation of variables

$$S = -Et + f_1(q_1) + f_2(q_2) + f_3(q_3)$$

(although may use any other method)